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TERM END EXAMINATIONS (TEE) – October 2026

Programme	:	**B.Tech.**	**Semester**	:	**Fall Semester 2026-27**
Course Name	:	**Database Management Systems**	**Course Code**	:	**CSE3001**
Date/ Session	:	**16-10-2026 / Session II**	**Slot**	:	**A14+D11+D12**
Time	:	**3 Hrs.**	**Max. Marks**	:	**100**

Answer ALL the Questions

Q. No.		Question Description	Marks
		PART A – (60 Marks)	
1	(a)	ABC Logistics is a growing company specializing in real-time package tracking. The company maintains a centralized Logistics Information System (LIS) to manage shipments. A ShipmentID uniquely identifies each Shipment and has attributes such as Weight, Dimensions, InsuranceAmount, Destination, and ExpectedDeliveryDate. Shipments are handled through Distribution Centres before reaching their final destination. Each Distribution Centre has a CenterID (Unique), Name, Location, and Capacity. To move packages, ABC Logistics utilizes different Transport Modes such as Air, Road, and Rail. Each Transport Mode has a ModeID, Type (Air, Road, Rail), and Capacity. A Shipment is associated with one or more Transport Events before reaching the final destination. An EventID uniquely identifies a Transport Event and includes details such as TransportType, ScheduleNumber, and RouteDetails. Each Shipment starts its journey from a Single Distribution Centre but may be transferred through multiple Transport Events before arriving at the final destination. Task: 1. Draw an Entity-Relationship Diagram (ERD) that represents the ABC Logistics tracking system. 2. Clearly indicate Primary Keys (Identifiers), Attributes, and Cardinality Constraints. Ensure many-to-many relationships (if any) are properly represented.	12
		OR	
	(b)	XYZ Group itself has up-to-date information on the processing and current location of each shipped item. To do this, XYZ relies on a company-wide information system. Shipped items are the heart of the XYZ product tracking information system. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date. Shipped items are received into the XYZ system at a single retail centre. Retail centres are characterized by their type, uniqueID, and address. Shipped items make their way to their destination via one or more standard XYZ transportation events (i.e., flights, truck deliveries). These transportation events are characterized by a unique scheduleNumber, a type (e.g, flight, truck), and a	12

		deliveryRoute. Create an Entity Relationship diagram that captures this information about the XYZ system. Be certain to indicate identifiers and cardinality constraints.	
2	(a)	Consider the following relational schema: Students (*student_id*, name, age, major), Enrollments (*student_id, course_id*, semester, grade), Courses (*course_id*, course_name, department), Instructors (*instructor_id*, name, department), Teaches (*instructor_id, course_id*, semester). Write the relational algebra expressions for each of the above queries. Clearly indicate selection (σ), projection (π), join (⋈), and set operations used in each query 1. Find the names of all students who are enrolled in the "Database Systems" course. 2. Find the names and majors of students enrolled in "Database Systems". Use a join between Students and Enrollments to get student details. The condition should match students to courses with the name "Database Systems". 3. Find the names and ages of students who are enrolled in "Database Systems" and have received an 'A' grade. Find all students who are taking at least one course taught by an instructor from their own department. Use Students, Enrollments, Courses, and Instructors tables. The selection condition should check if the student's major matches the instructor's department.	7
	(b)	*Discuss Codd's 12 rules for relational databases. Explain how modern RDBMS systems comply with these rules.* A database designer creates the following relational schema: Employee (EmpID, Name, Department, Salary, ManagerID) 1. Identify primary keys and foreign keys. Suggest improvements based on relational schema design principles.	5
		OR	
	(c)	For the given schema (the primary key of each relation is indicated by the underlined attribute), provide relational algebra expressions for the following queries: lives (person-name, street, city), works (person-name, company-name, salary), located-in (company-name, city), manages (person-name, manager-name) 1. Find the name of all employees (i.e., persons) who work for the City Bank company (which is a specific company in the database). 2. Find the name and city of all employees who work for City Bank. Similar to the previous query, except we have to access the Lives table to extract the employee's city. The join condition is the same person's name in the two tables, Lives and Works. 3. Find the name, street and city of all employees who work for City Bank and earn more than $10,000. Similar to the previous query, except for an additional condition on the salary attribute. Find all employees who live in the same city as the company they work for. For this query, we need to access the lives table to get the employee's city, the located_in table to get the company's city, and the works table to link the employee to their company. The selection condition is then that the two cities are the same.	12
3	(a)	What does SQL's Data Definition Language (DDL) serve as? Provide examples of CREATE, ALTER, and DROP statements.	6

	(b)	What are triggers in SQL? Provide an example where a trigger updates a "Logs" table. Explain the following TCL commands with examples: a) COMMIT b) ROLLBACK c) SAVEPOINT	6
		OR	
	(c)	Create a table called Employees with the following fields: • EmpID (Primary Key), Name (VARCHAR), Age (INTEGER), Salary (DECIMAL), DepartmentID (Foreign Key referencing Departments table) How do constraints like NOT NULL, UNIQUE, CHECK, and DEFAULT work in SQL? Provide examples.	12
4	(a)	Differentiate between a stored procedure and a function in PL/SQL. Write a PL/SQL procedure that takes an employee ID and displays their name and salary. Write a PL/SQL function that takes a number and returns its factorial.	12
		OR	
	(b)	Explain the different levels of RAID. How do they differ in terms of performance, redundancy, and cost? Compare RAID 0, RAID 1, and RAID 5 in terms of: • Data redundancy • Speed • Fault tolerance	12
5	(a)	Testing for serializability: Given the schedule below, determine if it is serializable: T1: R(A), W(A), R(B), W(B) T2: R(A), W(A) Construct a precedence graph and determine whether the schedule is conflict-serializable.	12
		OR	
	(b)	What is transaction isolation? Why is it necessary in concurrent execution of transactions? Describe the different isolation levels in SQL (READ UNCOMMITTED, READ COMMITTED, REPEATABLE READ, SERIALIZABLE).	6
	(c)	Define the concept of indexing methods and discuss the Sparse and Dense Index files with suitable examples.	6
		PART B – (40 Marks)	
6		As a DBA, you are given a poorly optimized database with redundant data. List five actions you would take to improve performance.	8
7		Explain multivalued dependencies and how they lead to Fourth Normal Form (4NF).	8
8		A database system requires high-speed access to frequently used data. Analyze whether magnetic disk or flash storage would be more suitable and justify your decision.	8
9		Write a PL/SQL block using an IF-ELSE statement to check if an employee's salary is above $50,000 and categorize them as "High Earner" or "Low Earner". Implement a FOR LOOP that prints the numbers from 1 to 10 in PL/SQL. Write a WHILE LOOP in PL/SQL that calculates the sum of the first 10 natural numbers.	8
10		Discuss concurrency control. Explain the two-phase locking protocol with an example.	8

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